UNIVERSITY OF BIRMINGHAM DUBAI

School of Computer Science

Human-Computer Interaction

FINAL DESIGN PROJECT

Interval — Schedule-Aware Fitness for University Students

Group 4

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File: HCI_FinalDesign_Group4.pdf

1. Team Manifesto

Each team member contributed actively across all phases of this project. Roles were rotated across interviews so that every member experienced conducting, note-taking, and observing. Prototype development was led by Mazen, with all design and evaluation decisions made collectively.

Member	Interview Roles	Key Contributions
Mazen Salama	Interviewer (P1), Note-taker (P4), Observer (P3)	prototype development; Stage 1 rewrite; design annotations; final compilation
Ayham Al Halabi	Interviewer (P2), Observer (P4), Note-taker (P3)	P2 interview; affinity mapping; problem statements,prototype development
Takwa	Interviewer (P3), Note-taker (P1), Observer (P2)	Persona development; user flow diagram;P3 interview,prototype development
Ali Abdelbadie	Interviewer (P4), Note-taker (P2), Observer (P1)	journey mapping; competitive analysis; P4 interview

2. Introduction

This report will document the complete process of the Human-Centered Design that was followed in the HCI Final Design Project at the University of Birmingham Dubai for the academic year 2025–26. The project was completed by four team members: Ali Abdelbadie, Ayham Al Halabi, Mazen Salama, and Takwa. Six different stages of the Human-Centered Design process were followed to develop the project: conducting research, conducting competitive analysis, conducting ideation, conducting prototype design, conducting usability testing, and making improvements to the project based upon the findings of the usability testing.

The research that was conducted for this project was built upon the findings from Assignment 1 for this course. In Assignment 1, four students from the university were interviewed regarding their reasons for not attending the gym. The major finding of this interview was that the students were not lacking motivation to attend the gym; their academic timetables did not permit for them to attend the gym. This finding informed the remaining stages of the Human-Centered Design process.

Interval is a fitness application that is designed for university students that have limited time to attend the gym after school. Interval works in a way that imports the student's timetable for the university, and automatically suggests gym attendances that will fit within their current schedule. Thus, Interval removes the difficulty that students encounter when attempting to find time to attend the gym with their classes and other academic commitments.

The target users of Interval are full-time students between the ages of 18 and 24 that are of similar profiles to those interviewed in Assignment 1: they are students that are motivated to attend the gym regularly but whose academic commitments prevent them from establishing a gym routine. Interval is designed to remove this preventative constraint upon these students.

Each of the six stages will be documented within this report. Stage 6, in particular, will explain the specific improvements that were made to the product following the evaluation of its usability, which was performed in Stage 5. Thus, each of the stages of the Human-Centered Design process are included in this report to provide an explanation of the reasons for each of the decisions that were made in the development of Interval.

3. Stage 1 — Understand the Problem & Users

3.1 Design Area

Our chosen design area is the achievement of fitness consistency among students who have limited time to dedicate to fitness. This topic was chosen for us due to its status as a genuine and widely-experienced problem. All four of our interview participants indicated that this was a real problem that they face in their lives.

3.2 User Research

To gather information regarding the experiences of students with fitness consistency, we performed four semi-structured interviews with university student participants (P1-P4). All of the interviews were conducted in person, and they took place between February 17th and 19th, 2026. The full documentation of these interviews, including their objectives, agendas, questions, and the interview notes, transcriptions, and summaries of the participants' responses and discussions is provided in Appendix B.

Semi-structured interviews were performed for this investigation to allow the participants to freely discuss the topics that they felt were important and relevant to the research questions, while also allowing the researchers to follow a pre-arranged interview guide to ensure that each interview covered the necessary topics. Each interview had one Interviewer, one Note-taker, and one Observer. The roles of each of these individuals were rotated after each interview. Verbal consent was obtained from each participant prior to any note-taking occurring during the interviews.

Code	Date	Interviewer	Est. Duration	Primary Focus
P1	19 Feb 2026	Mazen	~6min	Schedule constraints, morning preference, guilt spiral after skipping
P2	19 Feb 2026	Ayham	~3 min	Deadline unpredictability, exam season drop, mental health function
P3	19 Feb 2026	Takwa	~5 min	Full routine collapse, back-to-back lecture days, confidence impact
P4	17 Feb 2026	Ali	~3 min	Morning gym preference, early starts, outlier during exam season

3.3 Affinity Map

All interview notes were collected and converted to atomic sticky notes on Miro. Each sticky note contained one observation from the interview, and was labelled with the code of the participant who conducted the interview. These sticky notes were then grouped into clusters within Miro according to their themes and relationships with other observations from the interviews.

Cluster 1: Academic Time Conflict	Cluster 2: Psychological Factors	Cluster 3: Environment & Context
University schedule dictates gym time (P1-P4)	Guilt over spending time at the gym close to deadlines (P2)	Works out with scheduled gym classes (P1)
Exam periods cause the largest drop in gym visits (P1, P2)	Stress at the thought of skipping gym (P1, P2)	Preference for morning gym visits (P1, P4)
Cannot workout due to back-to-back classes (P3)	Cannot workout which impacts confidence to workout (P3)	Workout sessions are 45 minutes (P1) to 2 hours (P4)
University deadlines are not as predictable as other routines (P2)	students rarely use gym as an opportunity to skip studying (P2, P4)	Works out at 6-7pm five days a week (P2)

3.4 Persona

The following persona was synthesised from the interviews. This persona is representative of the typical student that participated in the interview and the student who is motivated to participate in physical activity but whose schedules prevent such activity from occurring.

Omar — Age 21 — Full-time University Student

Background

Second-year university student with a demanding timetable. However, he values his fitness and the way that it makes him feel mentally, yet his university schedule often prevents him from getting to the gym..

Goals for the student include wanting to:

- Maintaining a split of visiting the gym four times a week
- Avoiding long gaps between visits to the gym, especially during exam season
- Ensuring that gym visits can realistically fit in his academic schedule

Pain Points

- Timetable clashes with preferred workout times
- Early or long lecture days disrupt routine
- Unpredictable deadlines create routine instability
- Prep + unwinding make sessions feel longer than they are
- Exam season significantly increases avoidance

Representative Quote

"I want to stay consistent with visiting the gym, but during exam season and when my university schedule gets particularly busy, the gym is the first thing that I have to sacrifice."

Motivations • Feeling confident in his abilities to complete his university assignments • Relieve stress that he accumulates throughout the university semester	

3.5 User Journey Map

The journey map below traces Omar through a typical semester week, balancing his gym routine. Stage 7 illustrates the collapse of his gym routine during exam and midterm periods. Each of these pain points was gathered directly from the four participants of this study. 3.6 Key Insights

Stage	Actions	Thoughts	Pain Points	Emotions	Opportunity
1. Weekly Planning	Checks timetable; selects gym days around lectures	"If I plan around lectures, I can stay consistent."	Timetable controls gym timing (P1, P4)	Calm, cautious	HMW help students build a realistic gym plan around shifting schedules?
2. Uni Day Starts	Attends lectures; early or long days drain energy	"If today is packed, gym won't happen."	Early starts block gym (P4); long days disrupt (P1, P4)	Tired, frustrated	HMW support alternative timing when mornings/evenings are blocked?
3. Workload Increases	Assignments pile up; deadlines shift unexpectedly	"I'll go after I finish this... if I have time."	Sudden deadline changes (P2); group work disrupts (P2)	Rising stress	HMW reduce disruption from unpredictable workload changes?
4. Go/Skip Decision	Weighs gym vs study urgency; defaults to studying	"Studying feels more urgent right now."	Gym is first thing cut when busy (P2)	Torn, guilt building	HMW make the go/skip decision easier with minimum-viable options?
5. Workout Session	Goes to gym; attends scheduled class	"This helps my mindset and confidence."	Prep + unwinding inflate total time cost (P1)	Relief, improved mood	HMW reduce perceived time footprint of a session?

6. After Session / Skip	If skipped: studies more to compensate	"I need to make up for it by studying more."	Guilt spiral after skipping reinforces avoidance (P1)	Stressed, guilty	HMW prevent the guilt/stress spiral and help students re-plan?
7. Exam Period	Gym frequency drops sharply; study dominates	"Exams first. Gym can wait."	Frequency drops from 4–5x to once/week (P2)	Overwhel med, high stress	HMW keep exercise consistent during exams with low-friction routines?

3.6 Key Insights

The following insights were derived from the cross-participant analysis. These are interpretations of the data - statements of the reasons behind and implications of the design of the system and its components.

Insight 1

> The constraints of the university schedule are preventing students from showing up at the gym despite their desire to do so. Each of the four participants mentioned their desire to attend the gym but skipped sessions when they had full schedules. Consequently, the gym attendance system should be designed around the student's existing schedule rather than ask the student to change their habits.

Insight 2

> The decision of whether to go to the gym or to study is the highest friction point for the student since they almost always opt for studying. Reducing the friction at this point in the journey could significantly improve gym attendance.

Insight 3

> If students skip gym sessions, they often feel guilt and stress over it which makes it harder for them to attend gym sessions in the future. Student P1 mentioned that she felt compelled to study more to compensate for her skipped gym sessions. Thus, the system needs to account for the emotional consequence of skipping a gym session.

3.7 Design Goals

The following goals translate our key insights into specific, measurable outcomes the design must achieve:

G1	A user should be able to see their next available gym window and begin a matched workout in 3 taps or fewer from the gym application opening.
G2	The gym application should be able to recommend workouts based on a user's timetable.
G3	On days with heavy lecture schedules, a user should have at least one workout option that is 20 minutes or less.
G4	If a student skips a gym session, the application should reframe that gym session skip as a positive experience with the display of the next available gym session.

3.8 Design Challenge (HMW Statement)

> *"How might we help time-constrained university students to recognise and act on windows of time when they can undertake workouts within their timetable, without adding to the cognitive load that they already experience whilst students? "*

This question was chosen from a bank of questions that had been developed as a result of the research findings. The full bank of questions for consideration is displayed below.

Broad scope:

- How might we help university students to incorporate physical activity as a regular part of their academic life?
- How might we reduce the conflict that students feel between their academic commitments and their need to feel well?

Mid-range:

- How might we help time-constrained university students to identify windows of time within their timetable during which they can undertake workouts?
- How might we reduce the cognitive load that students feel when making the decision to undertake a workout that day?
- How might we prevent the feeling of guilt and stress that students feel when they do not workout that day?

Focused:

- How might we design an application that can recognise the student's next available time slot, and suggest a workout to be undertaken within that slot?
- How might we assist students who would like to undertake a physical activity, but who find it impossible to visit the gym that day?

4. Stage 2 — Competitive Analysis & Ideation

4.1 Competitive Analysis

We analysed three existing applications that address fitness planning for similar user groups — evaluating each against UX strengths, usability failures, and the design opportunity they present for Interval.

Nike Training Club

What works well	Aesthetic and minimalist design - allows for fewer visual elements to take the user's focus (Nielsen Heuristic 8). Uses a match between system and real world (H2) with exercise difficulty levels labelled as beginner, intermediate and advanced.
What fails	Fails on the flexibility and efficiency of use (H7) for the target user. The application assumes that the user has time to undertake the workouts yet does not provide any screen elements to indicate their availability to undertake these workouts. The main screen violates the principle of Gestalt proximity - elements that should belong together are not visually grouped.
Opportunity for Interval	To include the ability to view the user's available times to undertake workouts as a main feature of the application. Instead of asking the user what types of workouts they would like to undertake, the application can ask the user when they are available and only display workouts that fit around the user's availability.

MyFitnessPal

What works well	The visibility of the system status (H1) displays the users' caloric and macronutrient goals. The application implements recognition rather than recall (H6) in the database of names of foods and nutrients contained within them.
What fails	The application is entirely reactive to the user and only allows them to log the exercises that they have undertaken. Furthermore, the home screen presents the user with a high degree of cognitive load as it displays information regarding their caloric intake, macronutrients, steps taken, and exercise performed.
Opportunity for Interval	Replace the reactive nature of the application with one that suggests to the users whether or not they should undertake to undertake their set exercise goal at that specific time.

Freeletics

What works well	The personalisation of the workouts according to the user's performance and availability of time addresses one of the key pain points of one of our users. The app also incorporates coaching sessions following each workout to offer feedback and recovery information for the users.
What fails	The onboarding process for the app is quite challenging due to the profiling step that occurs prior to the user gaining access to any of the benefits of the service. Furthermore, the app assumes that the users have a workout schedule each week which is against the research findings about university students.
Opportunity for Interval	Designing an app that allows users to import their class timetable to the app will drastically reduce the amount of friction for the user to join the gym. The app will automate the scheduling process for the users which will eliminate the need for the users to plan their workouts.

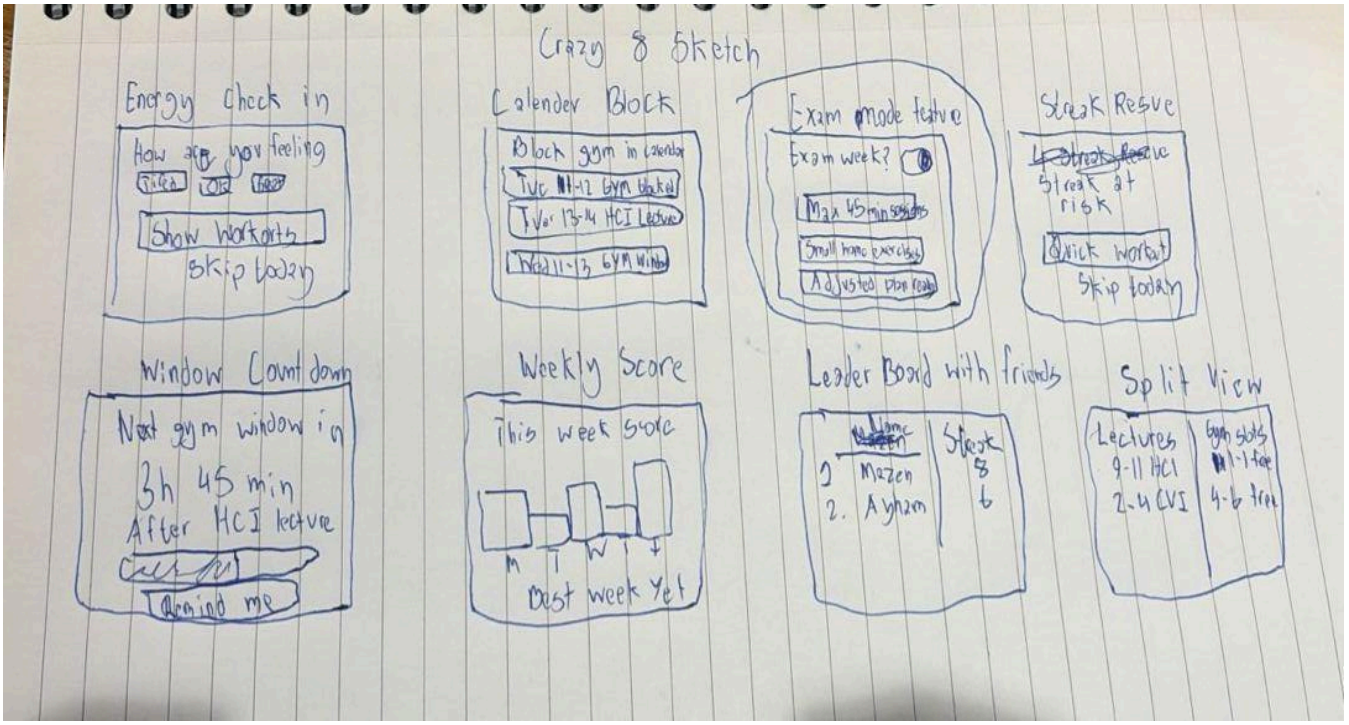
Hevy

What works well	The live timer for the workouts accurately displays the length of the workout when it is over. The saved routines feature allows the users to plan what types of workouts they will have at the gym when they arrive.
What fails	The app does not allow users to log in their workouts retrospectively. If they attempt to navigate to a previous date's workouts, a live timer will launch which is not the feature that users require after selecting a date for their workouts. Hevy does not have any scheduling or planning functionalities for the users. The calendar feature is only for retrospective logging of workouts.
Opportunity for Interval	Build a planning-first gym app that derives workout suggestions from a user's existing timetable to assist them in deciding when to go to the gym instead of simply having to decide what to do once they show up at the gym.

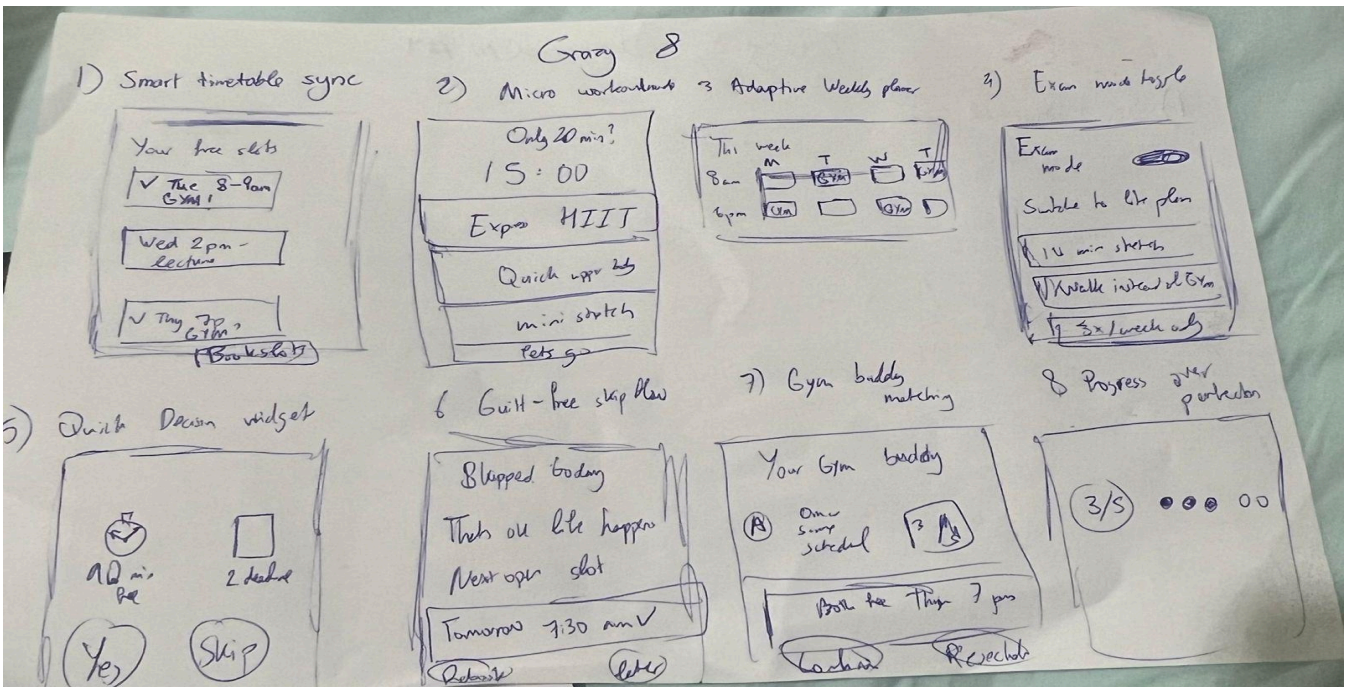
4.2 Ideation — Crazy 8s

Each member of the team independently ran a Crazy 8s ideation session. Each person was given a sheet of A4 paper that was to be folded into 8 panels. Each person had 8 minutes to complete their drawings where they were to design their ideas for the app. After the ideation session, each member shared their ideas and circled the strongest of their concepts

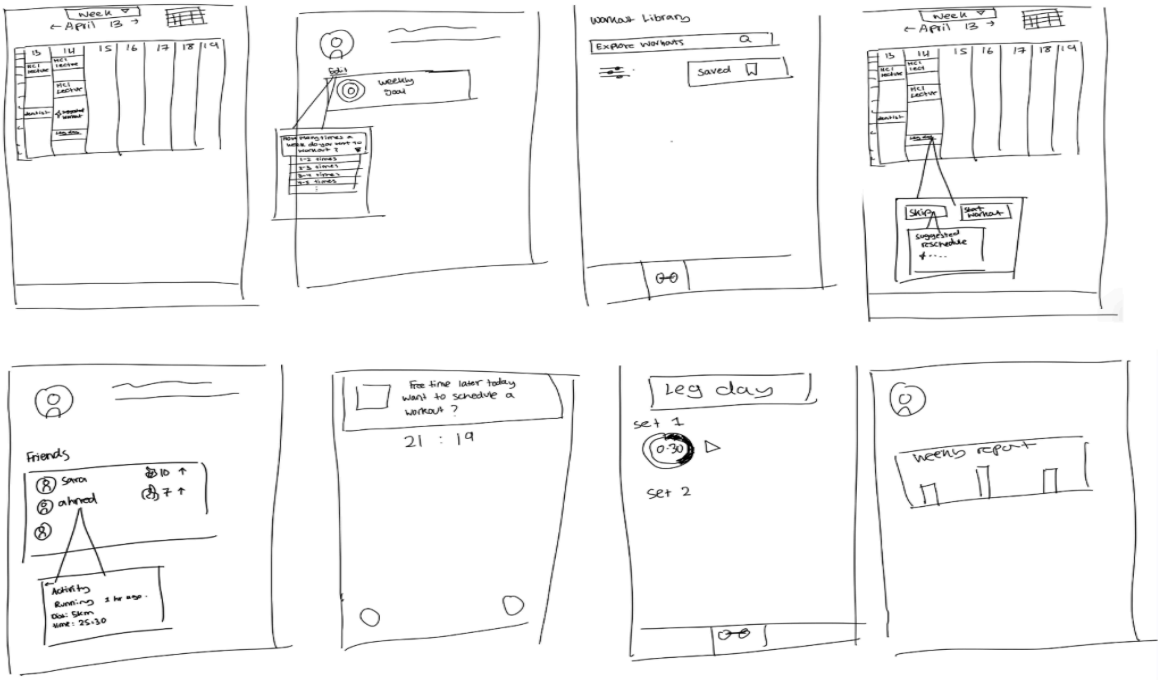
Mazen's Crazy 8s sketch:



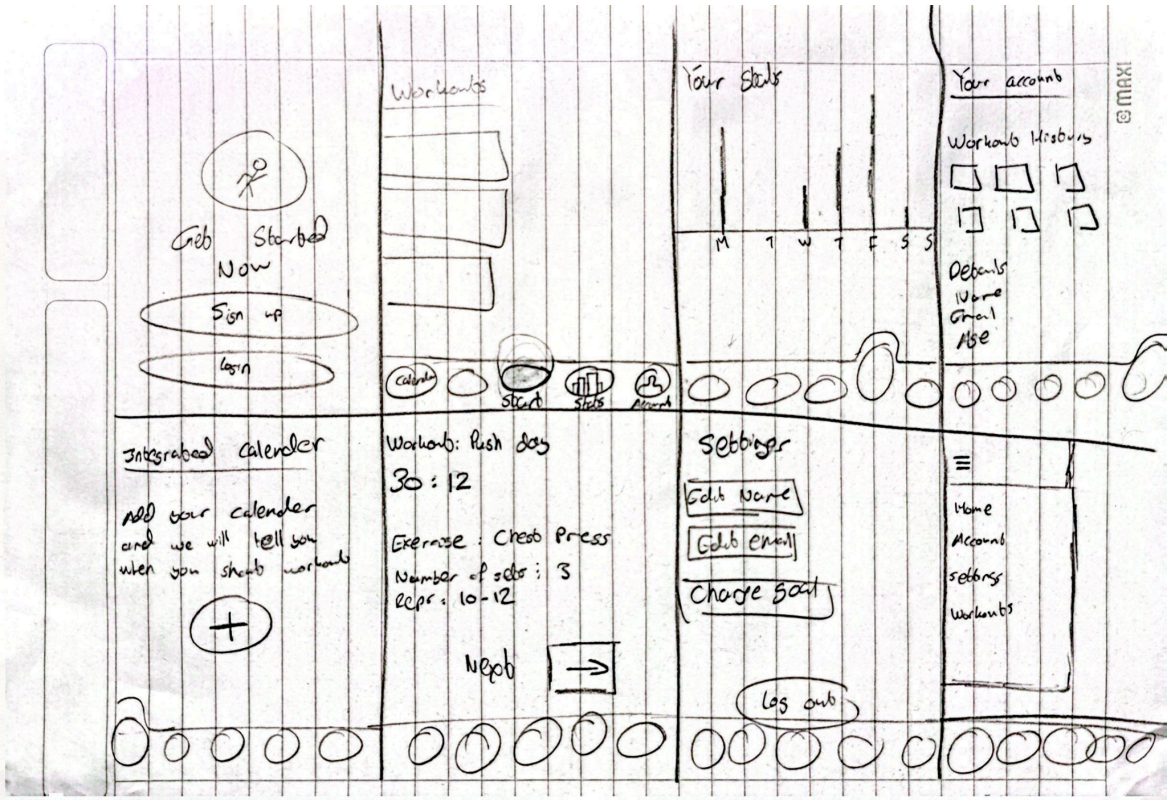
Ayham's Crazy 8s sketch:



Takwa's Crazy 8s sketches:



Ail's Crazy 8s sketch



4.3 Concept Selection & Rationale

Following the presentation of each member's strongest sketch, the group used dot voting to determine the leading concept for the project. Each member placed three votes for their top three ideas. Interval, the timetable-aware workout matcher, received the most votes due to the fact that it was the only idea to address each of the four design goals for which we gathered evidence through research.

Interval works to invert the typical model for fitness applications. Rather than asking the user for time to exercise, the program can utilize the user's timetable for the university to determine workouts that can be completed during their free windows. Thus, Interval addresses Insight 1 and Design Goal 2. Additionally, the app can provide micro-workouts to those with heavy lecture schedules (Design Goal 3), as well as allow for the user to skip workouts altogether (Design Goal 4). Thus, Interval is the only concept within ideation to have addressed all four design goals for the project.

Before selecting Interval, the group discussed the trade-offs between the four leading concepts that emerged from the Crazy 8s sessions. The gym buddy / social pairing concept (proposed by Ayham) scored highly on motivation but was deprioritised because it introduced a dependency on another user's availability, which conflicted directly with the scheduling flexibility that our research identified as the core need. The mood-based workout matcher (proposed by Mazen) was considered as a complementary feature rather than a standalone concept because it lacked the timetable-aware scheduling that all four design goals required. The micro-workout quick-launch concept (proposed by Takwa) addressed Design Goal 3 effectively but did not address the broader scheduling problem or the guilt-spiral identified in Insight 3. Interval was selected because it was the only concept that addressed all four design goals simultaneously and directly mapped to the research findings from all four participants. The timetable-import mechanism in particular was identified as the highest-impact differentiator, as no existing competitor app offered it.

5. Stage 3 — Design the Solution

5.1 Prototype Overview

Interval was prototyped as a fully interactive React application, live at: https://intervalhciproject.netlify.app/. The prototype covers 13 screens connected through a navigable user flow, with real interactions including a live workout timer, timetable preview, workout filters, and progress tracking.

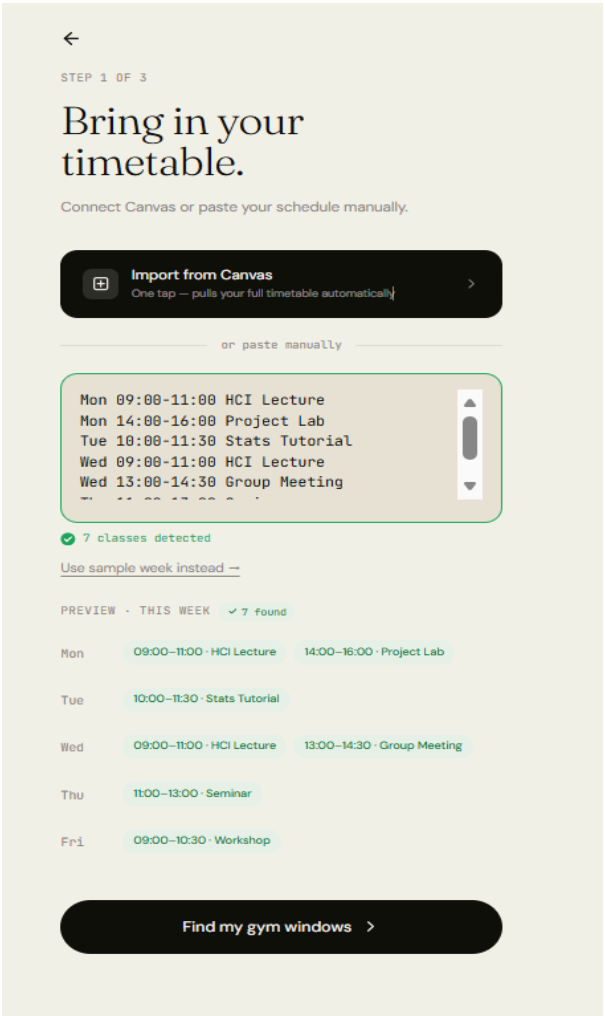
Tools used: React 18, Tailwind CSS, Lucide React icons, Vite build system. Typefaces: Fraunces (display), DM Sans (body), JetBrains Mono (data). All interactions are functional — not mocked.

Tools used: React 18, Tailwind CSS, Lucide React icons, Vite build system. Typefaces: Fraunces (display), DM Sans (body), JetBrains Mono (data). The prototype is coded rather than mocked — all interactions are functional.

5.2 Key Screens

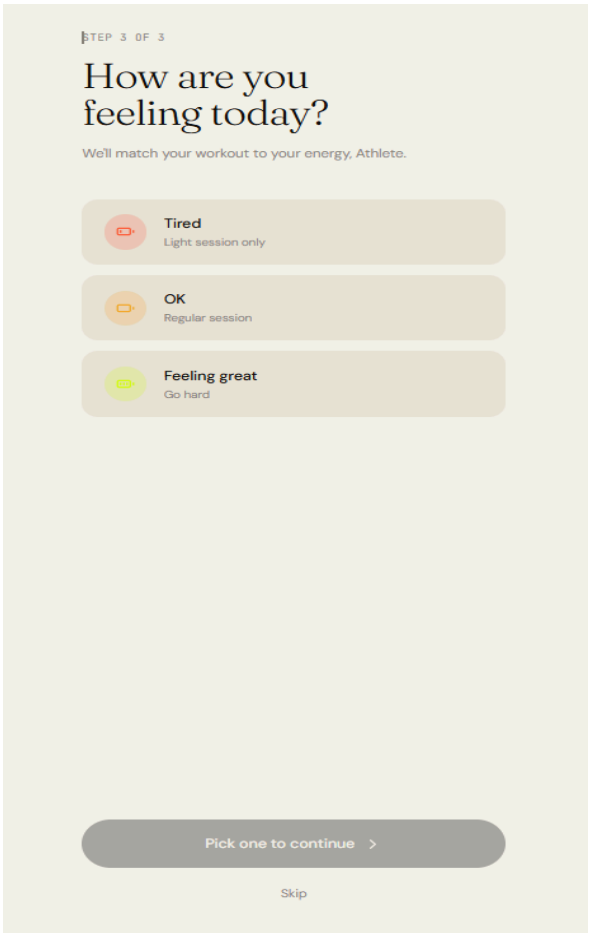
Screens 01–02 — Welcome & Schedule Setup:





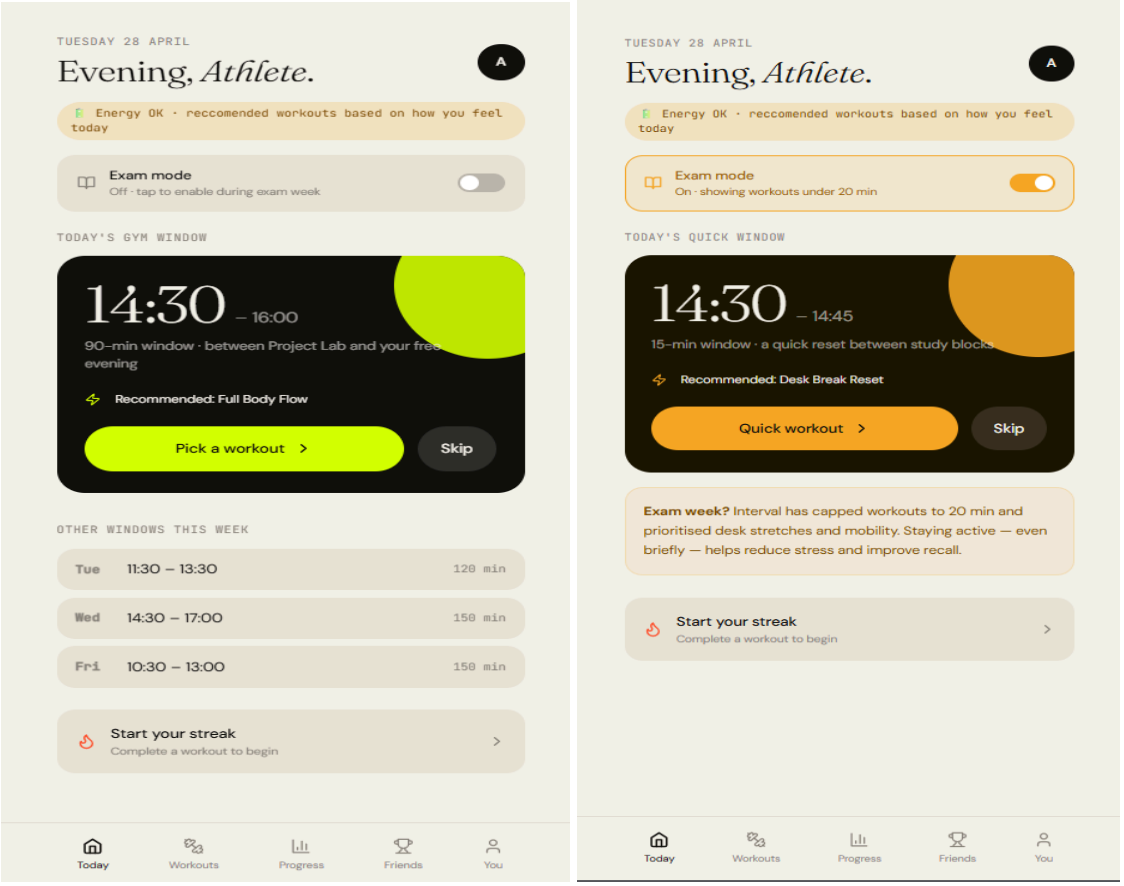
Screens 03–04 — Profile creation/User Sentiment:



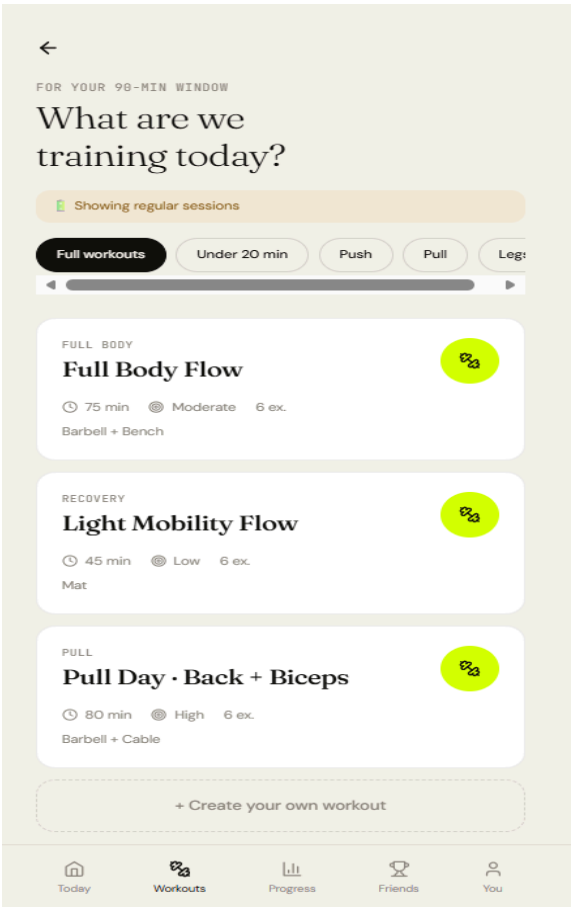


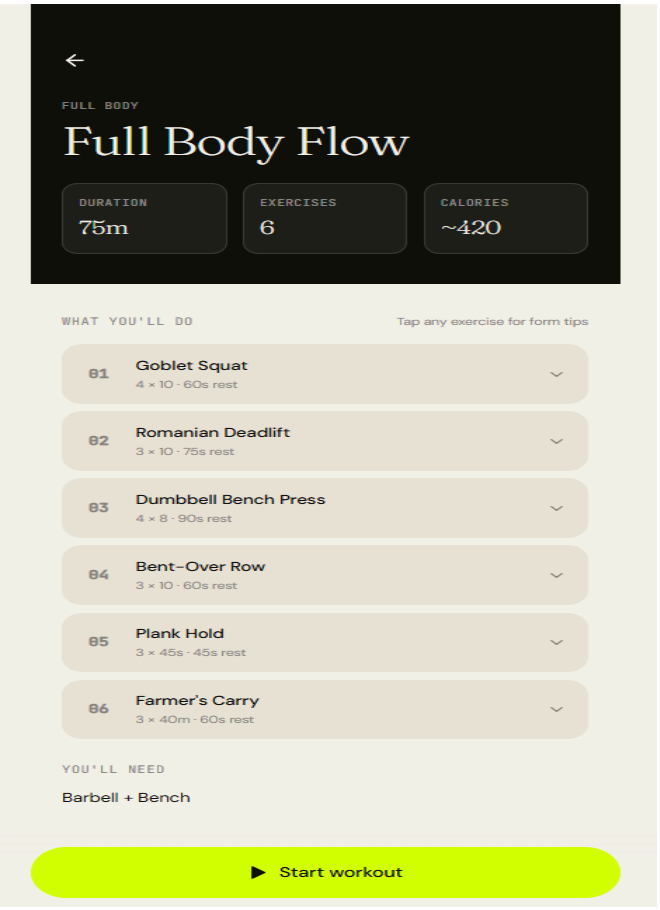
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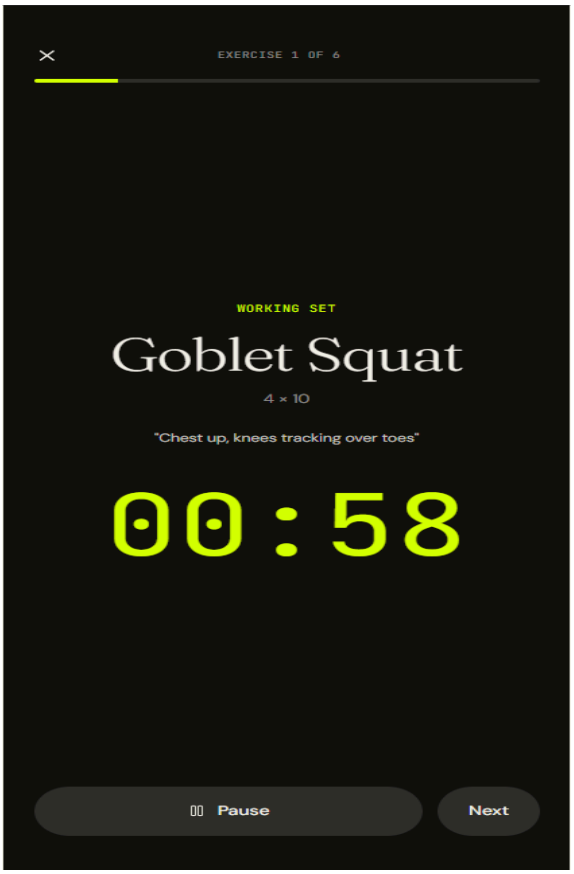
Screens 05–06 — Home Page(EXAM MODE (off/on)):

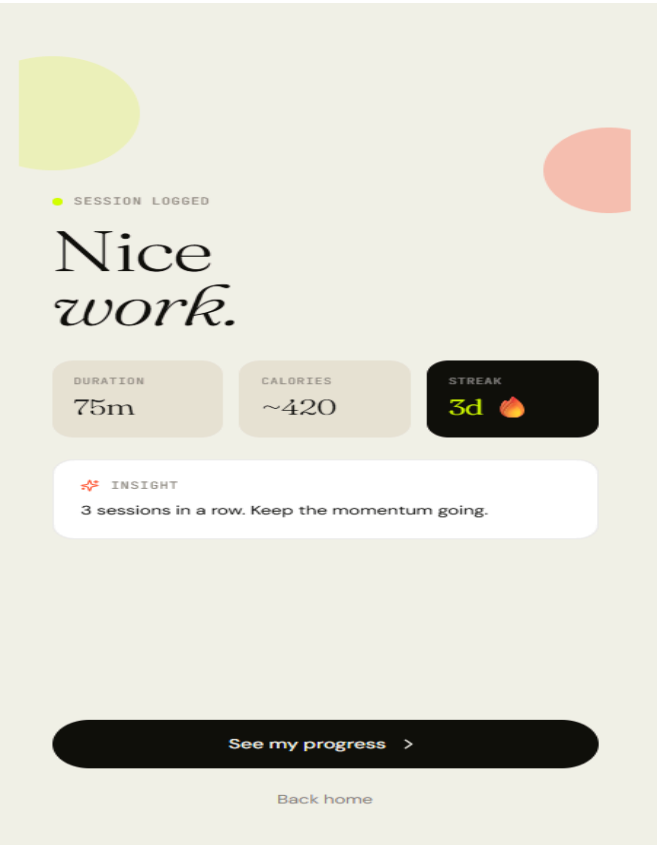


Screens 07–10 Workout flow:

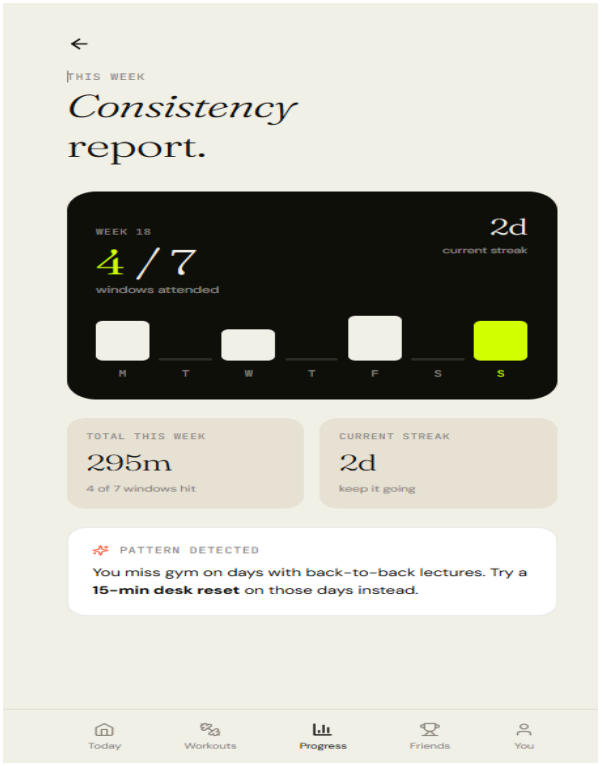


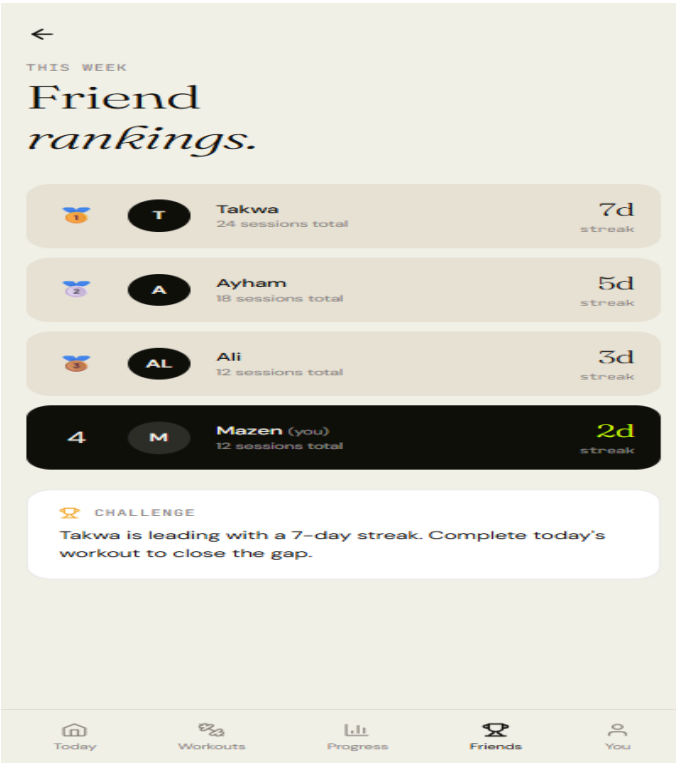


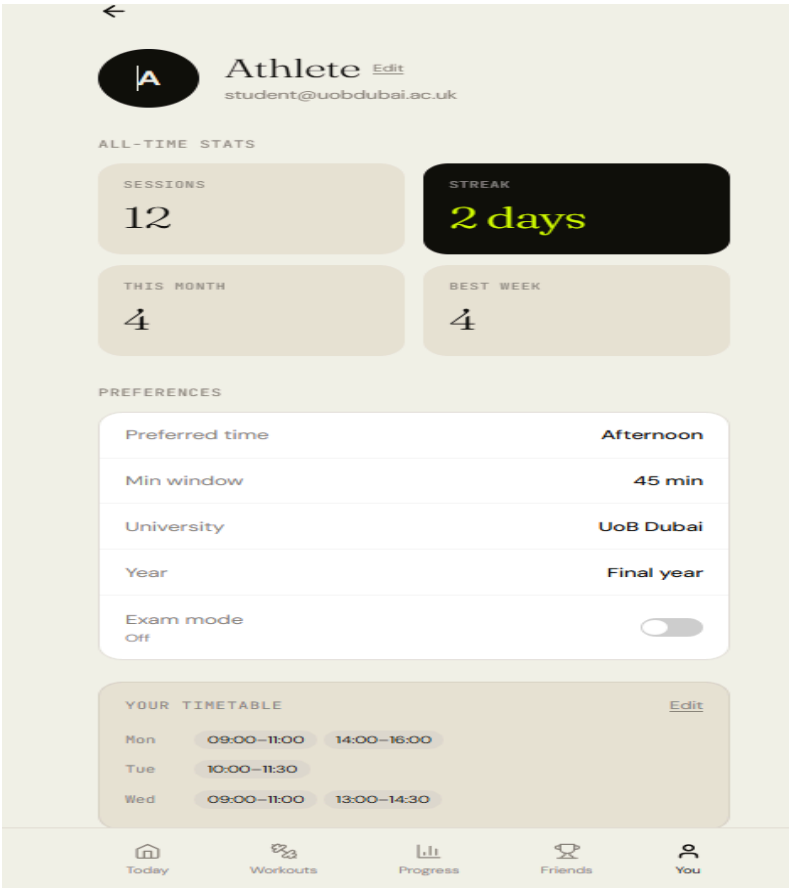




Screens 11–13 Progress, Friends, and Profile pages:







5.3 Design Annotations

The following annotations explain the key design decisions and the HCI principles that justified them:

Welcome Screen:

The editorial typography on the welcome screen ("Fit between classes") applies Nielsen's principle of matching the system to the real world (H2). Additionally, the single button on the screen applies the principle of aesthetic and minimalist design (H8).

Schedule Setup:

Making importing of the timetable the first interaction with the app before any workout content appears reflects the initial insight that guided the design of the app: schedule the workout first, then determine the content of the workout. The "use sample week" button applies error prevention (H5) by ensuring that even in the event of a failed timetable import, users can still interact with the application.

Home Dashboard:

The lime-coloured gym window card is the most visually dominant element on the home screen. This design decision applies the Gestalt principle of figure-ground to ensure that the most important element of the screen is also the most visually striking. The remaining gym windows apply the principle of visibility of system status (H1) - users can always view the schedule for the coming week.

Workout List:

The filter chips (“Full workouts” and “Under 20 min”) apply the principle of flexibility and efficiency of use (H7). For example, a student who has 90 minutes of free time available may have a different interest in the workouts compared to a student who can only devote 15 minutes to exercise. Additionally, the information that is grouped within each workout card also reflects the Gestalt principle of proximity.

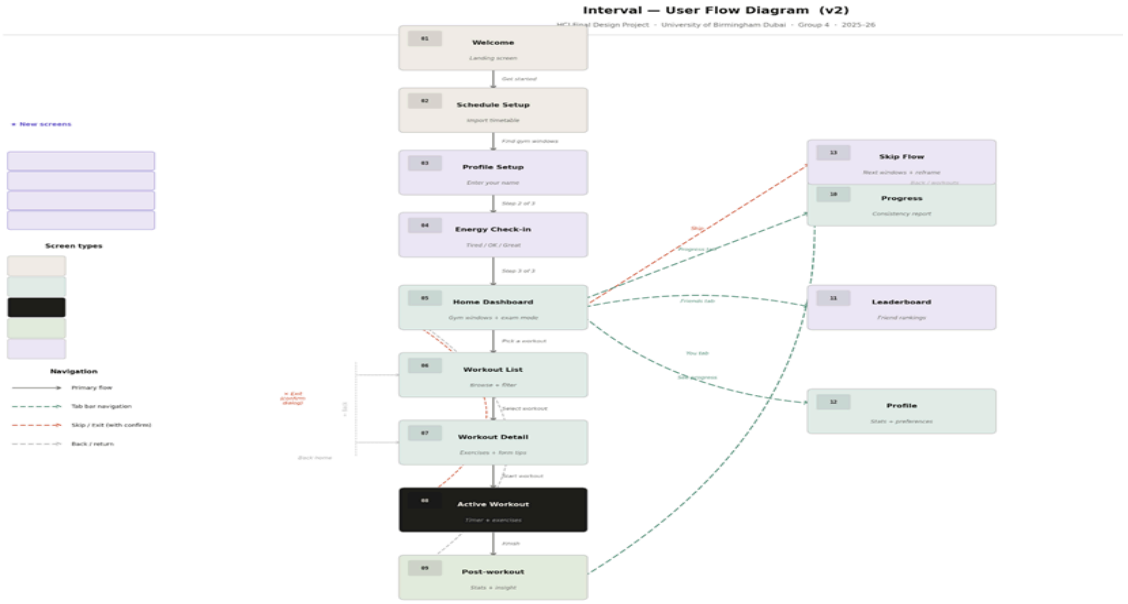
Active Workout:

The dark screen with the large workout timer applies the principle of visibility of system status (H1). Users can always see the timer to monitor their remaining time for the active workout. The visible exit button for the workout applies the principle of user control and freedom (H3). Furthermore, the progress bar reflecting the current exercise within the session indicates the status of the session.

Complete Screen:

The “Nice work” screen with insight into that just-completed workout session directly relates to Insight 3 regarding the guilt that students feel after completing a workout session. By providing positive reinforcement of the session that students have completed, this screen ensures that the design goal of reframing the experience after a session of exercise is achieved (G4)

5.4 User Flow Diagram



!

Colour palette:

Name	Hex Value	Role	Usage in UI
Cream	#F4F0E8	Background	All screen backgrounds; warm, low-fatigue base
Ink	#0F0F0E	Primary text	Body text, icon strokes, primary buttons
Lime	#D4FF00	Primary accent	Gym window cards, CTAs, active timer
Coral	#FF5733	Secondary accent	Streak indicators, insight highlights
Muted	#8A857C	Secondary text	Labels, timestamps, supporting information

Typography:

Typeface	Weight / Style	Usage

Fraunces	Light, Italic	Screen headlines and display titles (emotional, editorial)
DM Sans	Regular, Medium	Body text, labels, buttons, secondary information
JetBrains Mono	Regular	Times, data values, tags, and technical labels

6. Stage 4 — Evaluate the Design

6.1 Usability Testing

We tested the interactive prototype with participants who were not team members, using the live Netlify link. Each participant was given 3 tasks to complete without assistance. We observed task completion, recorded time on task with a stopwatch, and noted confusion points and direct quotes. The SUS questionnaire was administered after each session.

Tasks given to each participant:

1. Set up your timetable and find today's gym window
2. Pick a workout and start it
3. Find your weekly progress and streak

Task Completion Results

Participant	Task 1	Task 2	Task 3	Key observations
P1	✓ ~10s	Partial ~55s	✓ ~10s	Selected the "Pull" filter and noted no pull-specific workouts appeared. Also asked whether she could add her own custom workout rather than selecting from the app library.
P2	X ~25s	Partial ~53s	✓ ~8s	Got too lazy for fill in timetable suggested adding iway to import.Design liked and easiblity acknowledged; however, the exercise list lacking and recommended for exercises to have demo videos
P3	✓ ~25s	Partial ~43s	✓ ~5s	Tapped the × button during the Active Workout screen and lost all workout progress without any warning dialog. Frustrated at the app for the lost progress, she said “I thought it would ask me first.” She spent over 2 minutes on the Schedule Setup screen attempting to type in a timetable before selecting the sample week as an alternative. The screen featuring the “gym window” on the Home screen was somewhat unfamiliar to the user; she took a moment to read the screen to understand what was stated.

P4	✓ ~10s	Partial ~25s	✓ ~10	When selecting a timetable to import, the user chose to import the sample instead of selecting her own as it would have takes too long to type it manually. She was successful in selecting a workout and starting it without any issues. The weekly progress screen was accessed without struggling or visiting several pages to find the desired screen.

SUS Scores

participant	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	SUS score
P1	4	1	5	1	4	1	5	2	4	1	90
P2	4	2	4	1	4	2	4	1	4	2	**80.0**
P3	4	2	3	2	4	3	3	2	3	5	62.5
P4	4	1	5	1	4	1	4	1	5	2	90

Key Observations

All four participants indicated some form of difficulty on the Schedule Setup screen. Both P1 and P3 elected to use the sample week as opposed to inputting their own timetable. P3 spent more than 2 minutes on this screen, which was the longest time spent on any single task within any of the sessions.

The term “gym window” on the Home screen made all four participants hesitate. None of the participants were familiar with this term, and each took a moment to read it on the screen before continuing on with the session. For instance, P2 stated “does that just mean a free slot?” indicating that the users are likely to not be familiar with this terminology and it may need to be explained upon first use of the gym application.

P3 accidentally tapped the × button on the Active Workout screen and lost all of her progress on that session. This failure mode is the most serious of any of the identified issues. As a result of this failure, Improvement 2 (workout session save feature) was created.

P1 selected the Pull exercise in the Workout List but did not find any Pull exercises in the app. She asked if she could add a custom workout to the list instead. This indicates a gap in the app’s workout features as both limited variety and lack of an option to add custom exercises were both not accounted for by the designers of the application.

P2 and P4 were successful in completing all three tasks. As a result, P1 and P4 received the highest SUS score of 90 and P2 received the second highest score of 80. P2 stated that the exercise list was limited and that adding videos of each exercise would be beneficial. P1 agreed with this suggestion. Thus, each of the participants indicated that the limited content within the exercise application is the most often requested feature to be added to the app.

6.2 Heuristic Evaluation

Each member of the team evaluated the prototype using Nielsen's 10 Usability Heuristics. The findings of each team member were compiled in the table below (duplicates removed). Severity values are as follows: 0 = not a problem, 1 = cosmetic, 2 = minor, 3 = major,,4=critical.

Heuristic Violated	Issue	Screen	Severity	In Testing?	Suggested Fix
H7 — Flexibility & Efficiency	Timetable must be typed manually - no sync with Canvas/LMS means it takes 2+ minutes to enter and may be entered incorrectly	Schedule Setup	**3**	Yes	Add an "Import from Canvas" button that will auto-fill the timetable into the text area using the API that Canvas provides
H3 — User Control & Freedom	pressing the × button during an active workout will exit the app with no warning dialog and lose all workout progress for that session	Active Workout	4	Yes	Add a "confirm" dialog to the application that will appear when the user attempts to exit the workout session - such that the user must confirm which action they would like to take
H1 — Visibility of System Status	After typing in the text area on the Schedule Setup screen, there was no indication of whether the system had successfully received their data or not. There was no success indicator or live preview of the timetable after typing it in.	Schedule Setup	2	Yes	Show the user the preview of their created timetable within the application as they are typing it into the text area
H6 — Recognition over Recall	Exercise list shows names and cue text only — no demo video or image to show correct form Users must recall how to perform exercises from memory	Workout Detail	2	[Yes/ No]	Add a short video that demonstrates the proper form to perform each exercise within the application

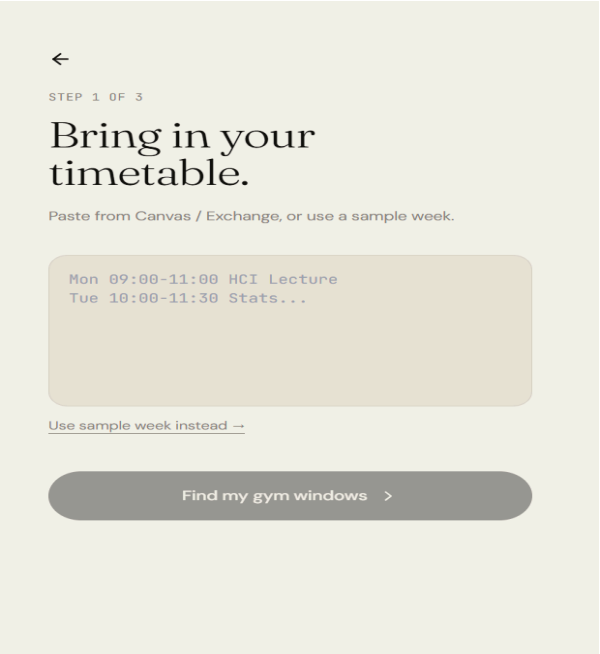
H5 — Error Prevention · H1 — Visibility of System Status	Filters showed unrelated workouts regardless of what was selected, giving users no feedback when a filter returned nothing — leaving them confused about whether the app had understood their choice	Workout list	3	yes	Ensure that filters to the workouts will only reveal workouts in the application that are correctly categorised - and that if a filter returns no results, the app will display a message to the user explaining the outcome of their filter
H3 — User Control & Freedom	The system did not allow for users to create their own workouts - only offering them a library of existing workouts to use for themselves.	Workout list	2	yes	Add a “Create your own workout” screen to the application where users can create a workout of their own - with their own exercises
H2 - Match between system and real world	Some of the content displayed was system-generated rather than natural language that could be understood by all users.	Home page and profile page	1	yes	Change the language within the application to that of any user can understand what is being displayed to them within the application. For instance: “0d” was changed to “0 days” within the application “workouts filtered” was changed to “recommended workouts based on how you feel today”

7. Stage 5 — Improve the Design

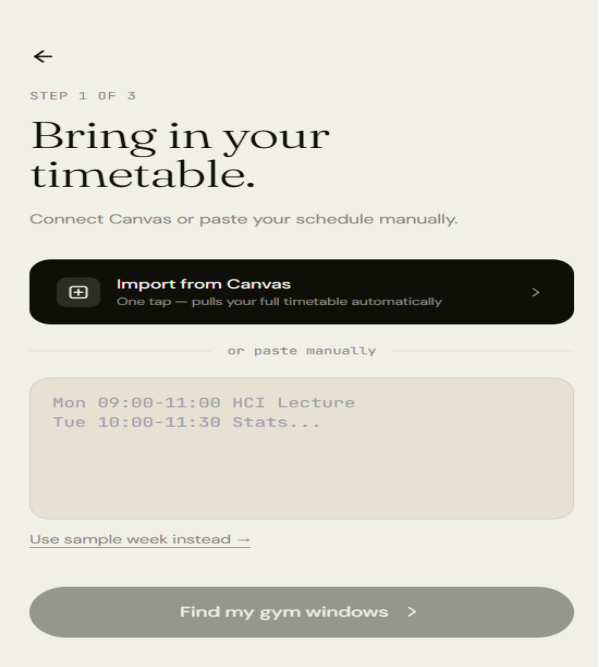
The issues were prioritised according to both the severity of the heuristic and whether they were seen during usability testing - an issue with severity of 3 but seen during usability testing will be prioritised before an issue of the same severity but not seen during usability testing.

7.1 Improvement 1

Before Screenshot:



After Screenshot:



Issue:

The Schedule Setup screen asks for users manually to type in their timetable. All three users got stuck on this screen. P1 and P3 both abandoned the schedule setup screen and used the sample class timetable. Furthermore, P3 spent over 2 minutes on this screen. There is no way for them to input their schedule, and there is no feedback for users if they have entered an incorrect schedule

Heuristic violated:

H7: Flexibility and Efficiency of Use. All of the users are required to manually type in their timetable - there is no way for them to skip to the next screen if they have their classes already uploaded into their Canvas account.

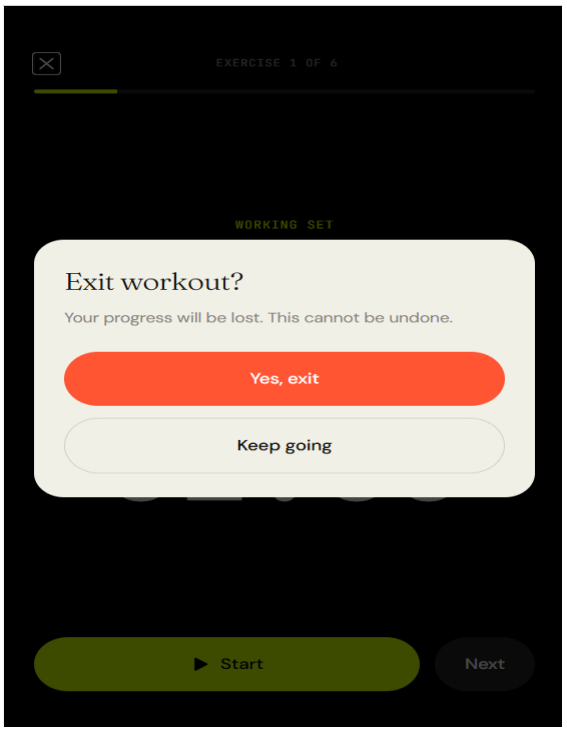
Change made:

An “Import from Canvas” button has been added to the Setup screen for users to automatically populate their classes into the application via the Canvas API - skipping the manual typing of the classes.

Evidence: In addition to the three participants who used the application, another developer also noticed this issue when reviewing the application. Furthermore, the severity of this issue is noted as 3 (major) and observed during usability testing. Finally, the fix was also independently made by Mazen to address this usability issue for the users.

7.2 Improvement 2

Afer Screenshot:



Issue:

When an user is actively performing a workout, tapping on the × button will exit them from the exercise - with no warning or confirmation. Furthermore, there is no way to recover their progress on that exercise session - it is instantly lost. This behavior indicates that there is an issue with the application's usability - particularly during usability testing.

Heuristic violated:

H3: User Control and Freedom. There is no way for an user to "cancel" the action of tapping on the × button - it will immediately and permanently lead to the Home screen for the user, without any means of recovery.

Change made:

A confirmation dialog will appear when the user taps on the × button - stating that "Exit workout? Your progress will be lost. This cannot be undone." The dialog will feature two buttons for the user to choose between - "Yes, exit" in coral red and "Keep going". Additionally, this

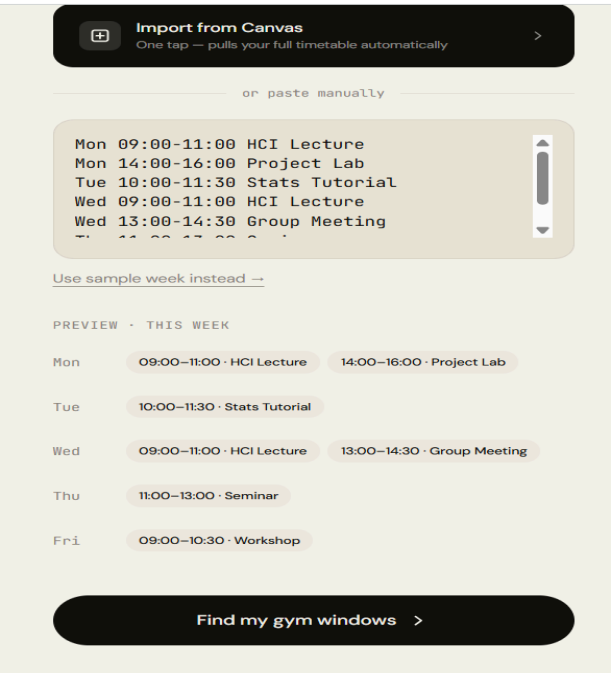
change has been made directly to the ActiveWorkout component of the application. The × button will no longer exit the user from the workout without confirmation.

Evidence:

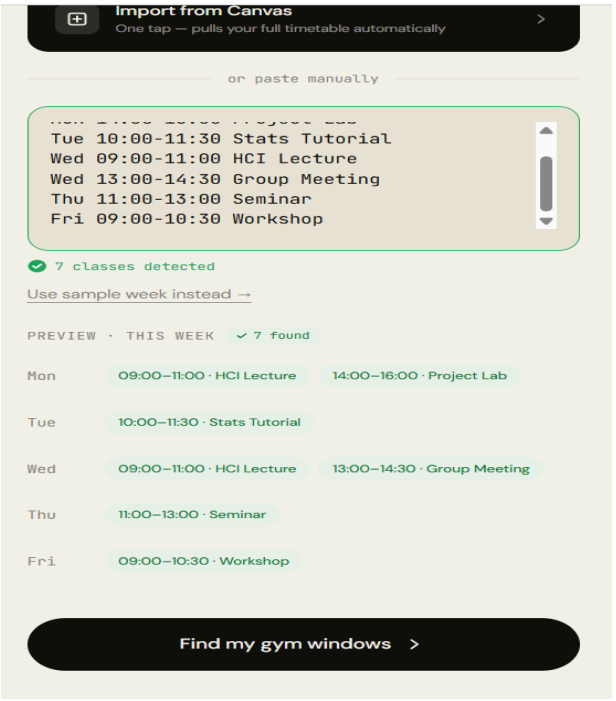
In addition to the three participants who used the application, another developer also noticed and created this fix for the application. For both participants, this issue was of severity 4 (critical) and was independently seen by both developers prior to usability testing.

7.3 Improvement 3

Before ScreenShot:



After Screenshot:



Issue:

When the user types in their classes into the text area on the Schedule Setup screen, there is no indication that the system has received that data. It is not until they reach the Home screen will they know that their data was not successfully populated into the application.

Heuristic violated:

H1: Visibility of System Status. The system does not provide any feedback to the user after the user types their classes into the text area - which is considered a violation of the heuristic.

Change made:

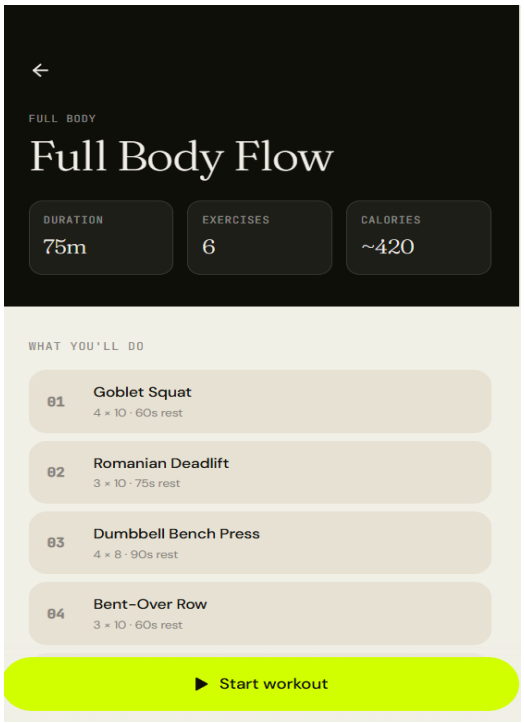
After the user types in their timetable, the application will automatically populate a live preview of the timetable that they created - again, without the user having to tap on a button. The display of the schedule will be accompanied by a green checkmark badge - indicating that the system recognized their number of classes. Additionally, if there are no classes defined within the text area, the preview will not appear for the user.

Evidence:

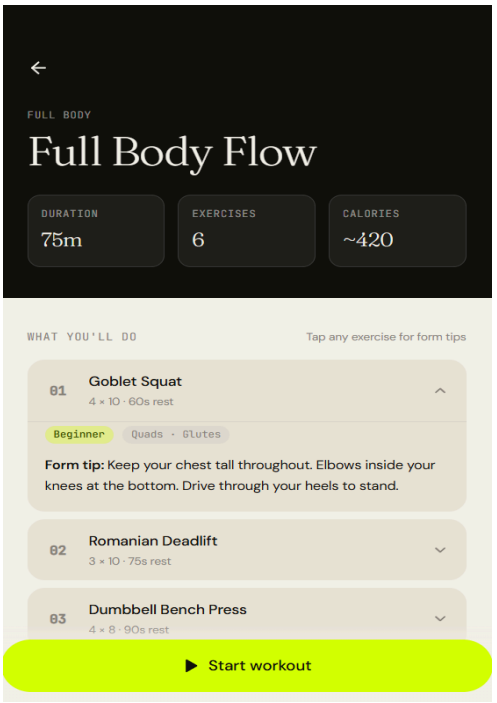
This issue was independently seen by Ayham - one of the developers who evaluated the application according to these heuristics. Furthermore, all four reviewers agreed to the severity - 2 being the highest - of this issue. Thus, this issue is considered to be a major usability issue - especially since it was also confirmed to exist during usability testing.

7.4 Improvement 4

Before Screenshot:



After ScreenShot:



Issue:

The Workout Detail screen does not include visual demonstrations of how to perform each exercise. For users who are unfamiliar with exercises such as Goblet Squat or Romanian Deadlift, there is no visual aid to help them understand how to complete the exercise. This was mentioned by P1 and P2 during usability testing.

Change made:

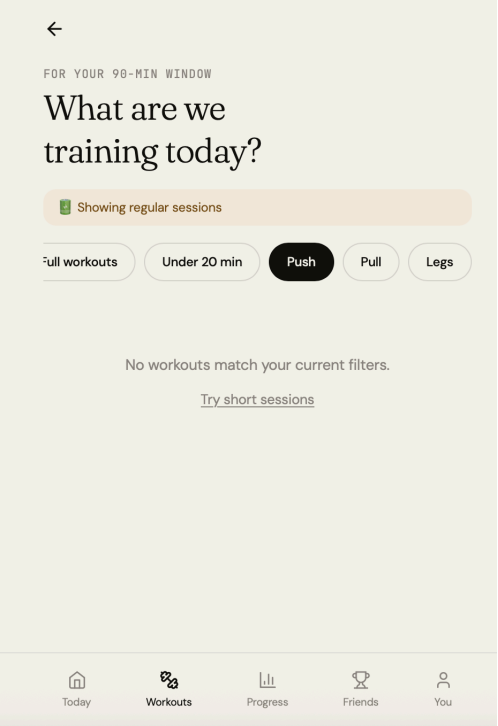
Each exercise on the Workout Detail screen now includes an expandable “form tip” section. Tapping on any exercise will reveal a difficulty level, muscle group that is targeted by the exercise, and the form that should be utilized when performing that exercise. The information is included directly within the exercise list so that users do not have to navigate away from the screen or rely on their memories of how to perform these exercises.

Evidence:

P1 and P2 both mentioned during usability testing that they were confused by the exercises and wondered how to perform them. Furthermore, the issue was independently flagged by Ayham during heuristic evaluation as having a severity of 2, and was confirmed through testing.

7.5 Improvement 5

After screenshot:



Issue :
When any of the Push, Pull, and Legs filter chips are tapped on the Workout List screen, the workouts that appear on the screen are not limited to only the exercises of that selected category. Additionally, if no workouts within that category are saved within the app, there is no message that indicates to users why the screen is displaying a list of exercises of only the categories listed in the app.

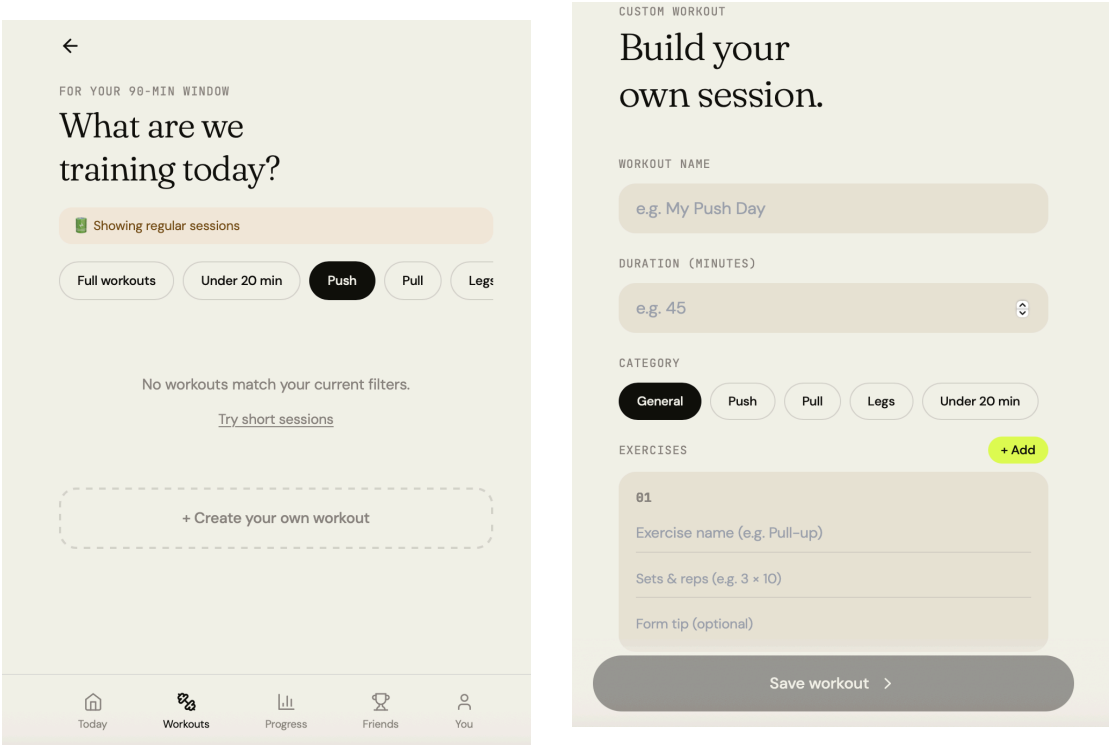
Heuristic violated:

Error Prevention and H1 — Visibility of System Status. The design allowed users to select filters that silently failed, with no feedback to indicate the issue or guide them toward a working option.

Change made: Change made: Each workout within the app now includes the category that the exercise belongs to. Additionally, if no workouts of that category are saved to the app, a message will display explaining that there are no workouts of that category saved so that users can select a different filter.

Evidence: During usability testing, P1 selected the Pull filter but no Pull exercises were saved within the app. Furthermore, this issue was flagged during heuristic evaluation with a severity of 3 which was also confirmed in usability testing to be the case.

7.6 Improvement 6



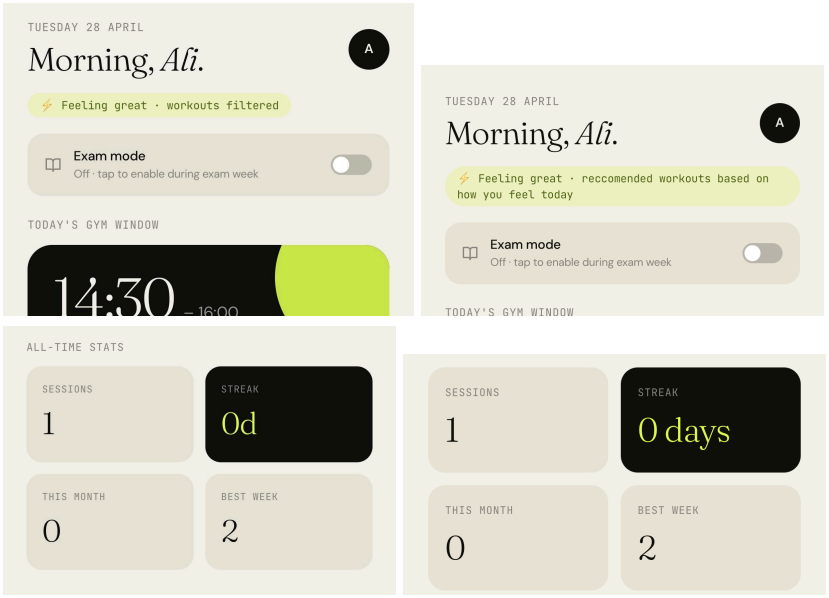
Issue:
There is no means for users to add their own workouts within the app. For users that have a routine that they use for their workouts, there is no means for the app to account for that routine or to allow users to build their own workouts.

Heuristic violated: H3 - User Control and Freedom
Change made: A “Create your own workout” screen has been added to the app. Users are able to add their own workouts by providing the name of the workout, the duration of the workout, the category that the users would like to associate with the workout, and their own list of exercises with sets, reps, and form tips. These workouts will appear within the Workout List screen in the same way as the other saved workouts.

Change made: A "Create your own workout" screen was added, accessible directly from the Workout List. Users can name their session, set a duration, choose a category so it appears under the correct filter, and add their own exercises with sets, reps, and optional form tips. Once saved, the custom workout appears alongside the preset options in the list and behaves like any other workout in the app.

Evidence: During usability testing, P1 expressed interest in the ability to create her own workouts. Furthermore, this issue was also flagged by Ayham during heuristic evaluation with a severity of 2, indicating the importance of providing such functionality for users within the app.

7.7 Improvement 7



Issue: Some of the wording implemented in the app may not be representative of natural language. For instance, the app displays “0d Streak” and “workouts filtered” throughout the application. Such wording indicates a potential problem between the system and the real world as the users may become confused by this language instead of understanding the meaning of the content of the app.

Heuristic violated: H2 - Match between system and real world

Changes made: Within the Workout List screen, the phrase “workouts filtered” was changed to “recommended workouts based on how you feel”. Furthermore, the “Streak” counter within the profile page was changed to read “days active” rather than the letters “d” and “o”.

Evidence: This issue was flagged by Ali when looking for things that could be improved in the app. It’s severity rating was only 1 but it was still dealt with to provide the best experience possible fo the user.

8. Stage 6 — Reflect on Learning

8.1 Individual Reflections

Each team member was asked to reflect personally on the HCD process — not to summarise the project, but to describe specific moments, unexpected discoveries, and genuine changes in thinking.

Mazen Salama

Honestly, the part of the HCI process I underestimated most was how much the research would constrain the design in a good way. Going through interviews, affinity mapping, and building the persona before touching any screens meant that when we got to ideation, we were not just guessing. We had a reason for almost every decision. That felt different from how I have approached design before, where I usually jumped straight into building something that seemed reasonable.

The most challenging stage for me was usability testing, and it was also the most important. I built most of the prototype, which made watching people use it uncomfortable. There was a moment when P3 tapped the × button mid-workout and the screen reset with no warning. She said "I thought it would ask me first" and I just had to sit there and take notes. I had looked at that screen probably two hundred times and never seen the problem. That is what I did not expect — building something and observing someone use it are not the same skill. You navigate based on what you know about how it works. They navigate based on what they can see.

What I would do differently is test much earlier, even with something rough. The issue with P3 would have appeared in week three if I had put a half-finished prototype in front of someone sooner.

Before this project, I did not necessarily know how to use heuristics as part of the design process. However, through this project I have come to understand the importance of making each of these decisions, even if they seem simple at first.

Ayham Al Halabi

The stage of the project that I reflect upon the most is the problem definition stage and specifically the process of narrowing down the problem that we were solving. This was arguably one of the hardest stages of the project. Each member of the team had somewhat different ideas of the problem that we were to solve.

My wrong assumption was that if we had more features to the product, then that was inherently a better product. I was focused on finding as many features as possible for the product to increase its value to potential customers.

The removal of each feature actually made the product concept stronger. This was a skill that I did not practice prior to this project, and one that I did not respect prior to this project.

The disagreements within the team were actually helpful; they led to some of the best design choices for the product. If I were to complete the project again, I would spend more time at the very beginning of the project focusing on the features that we should not include in our product. This would have helped to save the team from some of the backtracking that we had to take later in the project.

This project changed the way that I think about design. For me, design was seen as a creative project that was produced by the team. Now, I believe that design is a series of choices and the justification of each of those choices - a far more difficult and interesting problem than I had imagined going into the project.

Takwa

One of my main takeaways from this project is that users should be what generates the ideas, not the other way around. Before this I think my instinct would have been to try to come up with ideas I liked and then try to find who would use it. What I now realize is that by doing this you are essentially just designing based on your own assumptions and your perspective may not be representative. Starting from research and letting that drive the direction felt uncomfortable at first because it meant setting my opinions aside, but the decisions we made were stronger because of this.

Another thing I found very insightful was how important usability testing with real users turned out to be. When you have been working on something for weeks it is easy to become blinded to its flaws. Because you know how it works you stop being able to see where it does not. Seeing what our users observed and noticing the small things they hesitated on and the questions they asked helped us fill in those gaps. We thought we were pretty much done and we still came away with meaningful improvements that really brought the project together

If I did this again I would stick to less is more and just focus on the core insights we found rather than getting too caught up in features and ideas.

Ali Abdelbadie

One of the most important stages for me was the competitive analysis stage because it showed me that behind every great design, there is intention and a reason as to why the design is the way it is. Before this project, I usually thought designing an app meant creating something visually aesthetic with general functionality and useful features. However, when analyzing existing fitness apps, I realized that inspiration comes from understanding what works for the users and what frustrates them or causes inconveniences. What I found most valuable was learning to take inspiration from competitors instead of copying and building onto what already works to make the user’s experience better than what they are used to. Asking deeper questions as to why certain features work and others don’t helped us arrive at a solution that made sense for our target audience without overwhelming them with unnecessary features that add no value to their lives. The most surprising part was realizing that even popular apps contain usability issues. This showed me that no design is perfect and that improving an already working app is a never-ending process even when you have a lot of satisfied users, there is also something to improve. if I was to repeat this project, I would look more into the psychology of the target audience and try to understand

exactly what the users want and how I can provide everything for them in a way that benefits them and me as well. This project made me look at designing digital products in a different way. Back then I thought it was giving the user something they want and making it look nice, but now, I understand that it is more than that and that there are more levels to it than I thought.

8.2 Group Synthesis

When we all got together after writing our thoughts the first thing that hit us was that none of us had talked to each other before writing them. We had all gone off on our own. Come back with different stories about different moments. And yet we kept seeing the same things over and over.

The one thing that really stood out was that we all started this project with ideas we did not even know we had and working on it changed all of those ideas. Ayham thought that more features would make a product. Ali thought that good design was about how something looked and worked. Takwa thought that you start with ideas and then find people to try them out. Mazen thought that if a design made sense to the person who made it it would make sense to the person using it.. None of those ideas lasted when we actually talked to real people. What really got us was that we did not change our minds because we talked about it. We changed our minds because of what people told us. Someone said "I thought it would ask me first." We did four interviews. Found out that peoples schedules were the problem, not their motivation. We had filter chips that did not work and did not tell us anything was wrong. The Human Centered Design process makes us see these things. That is what makes it so useful. You cannot just argue your way out of being wrong. You have to see it for yourself.

The second thing that came up was the problem with moving fast. Takwa thought that we still had ideas when we were brainstorming. Mazen made a screen that did not work when we tested it.. We could have caught that problem earlier. Ayham said that we had to go and fix things because we did not ask the right questions soon enough. In each case the problem was the same: we moved on to the step before we were ready. If we had taken our time at the start we would have saved time later. That seems weird when you are trying to meet a deadline. It is what we should have done.

The third thing was about not doing much. Both Ayham and Takwa said that less is more and Ali found out that even popular apps have problems and nothing is ever really finished. Good design is not about doing things. It is about making choices. Every time we removed a feature Interval got better. Every time we used research instead of guessing, our design got stronger.

If we learned one thing from this project it would be this: take the research part seriously. It is the important part. Everything else whether it is the people we are designing for, the design goals, the prototype, it all changes. The research is the most crucial part.

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